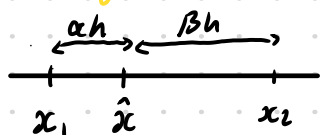


Taylor's Theorem

If the function $f(x)$ is differentiable at least $n+1$ at a point $x=a$, then values of f near to a are given by:

$$f(a+h) = \sum_{r=1}^{n+1} \frac{h^r \frac{d^r f}{dx^r}}{r!} + O(h^{n+1}) \quad \begin{array}{l} \text{Big O notation 'O(n)'} \\ \Rightarrow \text{tends to 0} \end{array}$$

Convergence of linear interpolation



$$\begin{aligned} \hat{x} - x_1 &= \alpha h \\ x_2 - \hat{x} &= \beta h \end{aligned} \quad \rightarrow \quad \boxed{\alpha + \beta = 1}$$

$$\hat{f}(\hat{x}) = \frac{(\hat{x} - x_1)f(x_2) + (x_2 - \hat{x})f(x_1)}{x_2 - x_1}$$

$$= \frac{(\alpha h)f(\hat{x} + \beta h) + \beta h f(\hat{x} - \alpha h)}{x_2 - x_1}$$

$$= \alpha f(\hat{x} + \beta h) + \beta f(\hat{x} - \alpha h)$$

$$\approx \alpha \left[f(\hat{x}) + \beta h f'(\hat{x}) + \frac{\beta^2 h^2}{2} f''(\hat{x}) \right] + \beta \left[f(\hat{x}) - \alpha h f'(\hat{x}) + \frac{\alpha^2 h^2}{2} f''(\hat{x}) \right]$$

$$\hookrightarrow (\alpha + \beta)f(\hat{x}) + \frac{\alpha \beta h^2}{2} (\alpha + \beta) f''(\hat{x})$$

$$\Rightarrow f(\hat{x}) + \frac{\alpha \beta h^2}{2} f''(x)$$

$$\text{Error in approximation} = \epsilon = |\hat{f}(\hat{x}) - f(x)|$$

$$= \left| \frac{\alpha \beta h^2}{2} (\alpha + \beta) f''(x) \right| = O(h^2)$$

When θ is small, $\sin \theta \approx \theta$

$$\cos \theta \approx 1 - \frac{\theta^2}{2}$$

$$\tan \theta \approx \theta$$